TKS Cognitive Framework

Referee-Safe Reduction and Claim Classification

Prepared from the December 2025 source text

April 8, 2026

Abstract

This document is a referee-safe reduction of the source file `TKS_FULL_Cognitive_Framework.tex`. Its purpose is editorial rather than promotional: it preserves only those statements that can be honestly presented as definitions, adopted axioms, imported standard mathematics, explicit threat models, heuristics, or short proved consequences of stated definitions. Claims of empirical validity, predictive power, psychological diagnosis, population-level control, or resolved AI safety are removed or reclassified as unsupported application hypotheses. The result is not a proof of the source document's psychological, social, or AI claims. It is a narrower artifact designed to separate formal structure from interpretation and to mark clearly where validation is still missing.

1 Scope and Labeling Rule

Definition: Editorial rule

Every substantive retained claim in this reduction is labeled as exactly one of: **Definition**, **Axiom**, **Imported Standard Result**, **Threat Model**, **Heuristic**, **Application Hypothesis**, or **Proved Statement**.

Definition: Referee-safe meaning

"Referee-safe" here means only that the claim surface is honest and bounded. It does *not* mean that the underlying framework has been empirically validated.

Threat Model: Intended use

Material retained below for psychology, social influence, and predictive analysis is presented only as a defensive or analytic threat-model language. Nothing in this document should be read as an endorsed deployment recipe.

2 Formal Core Retained From the Source

Definition: Noetic alphabet

Adopt a finite alphabet of ten labeled operators

$$\mathcal{N} = \{\nu_0, \nu_1, \ldots, \nu_9\}.$$

In this reduction, the operators are treated as named symbols inside a formal model. No empirical semantics are assumed beyond the labels chosen by the framework.

Axiom: D/W/P acquisition order

For each foundation index m, the acquisition predicates satisfy

$$\sigma(P_m) \Rightarrow \sigma(W_m) \Rightarrow \sigma(D_m).$$

This is retained as a proposed structural axiom of the framework, not as an empirical law of human cognition.

Definition: Chain level

For each foundation m, define a chain level by

$$\mathrm{Level}_m = \begin{cases} 0 & \text{if } \neg\sigma(D_m), \\ 1 & \text{if } \sigma(D_m) \wedge \neg\sigma(W_m), \\ 2 & \text{if } \sigma(W_m) \wedge \neg\sigma(P_m), \\ 3 & \text{if } \sigma(P_m). \end{cases}$$

Definition: Profile sequence

A profile sequence is a finite or countable word

$$\Phi_{\text{profile}} = \langle k_1 : k_2 : \cdots : k_n \rangle, \qquad k_i \in \{0, \ldots, 9\}.$$

This is a formal encoding device only. No automatic extraction validity is claimed here.

Definition: Grouped domain scores

If a positive 40-state vector $x = (x_1, \ldots, x_{40})$ satisfies $\sum_{i=1}^{40} x_i = 1$, define four grouped domain scores

$$d_j(x) = \sum_{i=10(j-1)+1}^{10j} x_i, \qquad j = 1, 2, 3, 4.$$

Proposition 1 (Grouped simplex property)**.** *If $x_i > 0$ for all i and $\sum_i x_i = 1$, then each $d_j(x) > 0$ and $\sum_{j=1}^{4} d_j(x) = 1$.*

Proof. Each $d_j(x)$ is a sum of positive terms, so $d_j(x) > 0$. Summing the four block sums simply

partitions the total sum of the forty coordinates, hence

$$\sum_{j=1}^{4} d_j(x) = \sum_{i=1}^{40} x_i = 1.$$

□

Proposition 2 (Immediate consequence of the D/W/P axiom)**.** *Under the retained acquisition axiom,* $\mathrm{Level}_m = 3$ *implies* $\mathrm{Level}_m \geq 2$*, and* $\mathrm{Level}_m \geq 2$ *implies* $\mathrm{Level}_m \geq 1$*.*

Proof. If $\mathrm{Level}_m = 3$, then $\sigma(P_m)$ holds, hence by the axiom $\sigma(W_m)$ and $\sigma(D_m)$ also hold. Therefore the chain has at least the properties required for the lower levels. The second implication is identical. □

3 Imported Mathematical Templates

Imported Standard Result: Contractive fixed-point principle

If a self-map on a complete metric space is contractive, then it has a unique fixed point and iterates converge to it. This is standard mathematics. It does *not* by itself validate any psychological or social interpretation.

Definition: Lacunarity template

For a suitable box-mass function M_ε, one may define

$$\Lambda(\Phi, \varepsilon) = \frac{\mathrm{Var}[M_\varepsilon]}{\mathbb{E}[M_\varepsilon]^2} + 1.$$

This is a mathematically standard kind of gap statistic in fractal analysis.

Imported Standard Result: Sheaf language

If a site $(\mathcal{C}, J)$ is given, then the category $\mathbf{Sh}(\mathcal{C}, J)$ of sheaves is a Grothendieck topos. This is standard category theory. Any use of sheaf language for values or cognition requires additional modeling choices.

Imported Standard Result: Graph-based threshold dynamics

Threshold and cascade models on graphs are a standard mathematical template for studying diffusion processes. Their use in any concrete social or cognitive application depends on justified state variables, parameters, and data.

4 Retained Threat Models and Heuristics

Heuristic: NLP-to-noetic mapping

One may attempt to map text or behavior into profile sequences Φ_{profile} by a classifier over the ten-symbol alphabet. In this reduction, that mapping is treated as a heuristic feature-extraction step. No claim is made that such a mapping is unique, validated, or psychologically complete.

Threat Model: Psychological profiling

Given a chosen extraction pipeline, one can define a workflow

$$\text{text/behavior} \longrightarrow \Phi_{\text{profile}} \longrightarrow \text{derived features}$$

for defensive analysis of possible profiling systems. This is retained only as a threat model describing how an adversary might operationalize a symbolic framework.

Heuristic: Attractor language for profiles

If a profile update map is *assumed* to be contractive on a chosen state space, then fixed-point language may be used to speak about an attractor-like summary state. In the absence of a validated state space and validated dynamics, this remains heuristic rather than diagnostic science.

Heuristic: Vulnerability score

The source document's vulnerability score based on lacunarity is retained only as a heuristic ranking function. Thresholds such as "critical if $\mathcal{V} > 1.5$" are not preserved as validated scientific cutoffs.

Threat Model: Influence and cascade modeling

Graph diffusion, seed-node choice, campaign propagation, and attractor perturbation can be written down as formal threat models for how a hostile actor might reason about influence operations. They are not presented here as proven laws of populations.

Threat Model: Predictive pipeline

The six-stage pipeline from digital traces to inferred goals, trajectories, and intervention points is retained only as an adversarial capability model. It is not retained as an established predictive system.

5 Application Hypotheses for AI

Application Hypothesis: Alignment decomposition

The source proposes that alignment analysis may be *modeled* by decomposing goals into desire, wisdom, and power components and expressing constraints through an RPM-like wrapper. This is a design hypothesis, not a solved alignment theorem.

Application Hypothesis: Interpretability vocabulary

The ten-symbol noetic alphabet may be explored as a human-readable vocabulary for annotating model behavior. This is a proposed interpretive schema, not an established decomposition of neural computation.

Application Hypothesis: Value-sheaf viewpoint

It may be mathematically convenient to describe context-dependent value assignments using sheaf-like language over a chosen site. This is a modeling proposal only; no result here shows that real value learning in AI is solved by such a construction.

Application Hypothesis: Safety architecture

Temporal typing, coalgebraic dynamics, or categorical organization may be useful for specifying AI safety architectures. The source document does not provide the formal verification machinery or empirical evidence needed to claim deployment readiness.

6 Claims Explicitly Removed From the Referee-Safe Version

Definition: Removed claim class

The following kinds of claims from the source document are intentionally omitted from the referee-safe version because they were unsupported, overstated, or too operational for a scholarly artifact.

- theorem-style claims without proofs, including the "40-Step Stabilization" claim and application-level cascade or predictive theorems;
- assertions that TKS provides "rigorous solutions" to core AI crises;
- statements that collective attractors, value resolution, or uncertainty tracking make AI systems robust or safe "by construction";
- operational scenario language that reads like a deployment recipe for psychological manipulation, population steering, or mass surveillance;
- threshold claims presented as if empirically calibrated when no evidence is supplied.

7 What Would Be Required for Validation

Definition: Validation requirements

To move any retained heuristic or application hypothesis toward scientific credibility, the framework would need at minimum:

(1) an explicit state space and observation model;

(2) reproducible extraction procedures from data to symbolic states;

(3) benchmark datasets and inter-rater agreement where human annotation is involved;

(4) predictive evaluation against held-out outcomes;

(5) ablations against simpler baselines;

(6) false-positive and false-negative analysis for any risk or vulnerability scoring.

8 Conclusion

Definition: Honest status

This referee-safe reduction does *not* certify the original cognitive framework as scientifically valid. It preserves a narrower core: formal definitions, a few trivial proved consequences, standard imported mathematics, and clearly labeled heuristics or threat models. Everything stronger remains future work or unsupported interpretation.

References

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[3] S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic*, Springer, 1992.

[4] D. Centola and M. Macy, "Complex contagions and the weakness of long ties," *American Journal of Sociology* 113(3) (2007), 702–734.